The Discrete-Time Fourier Transform

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Key Concepts

1) The discrete-time Fourier transform (DTFT) is used to analyze discrete-time signals and systems.

2) The DTFT applies to discrete-time non periodic signals, but can be extended to discrete-time periodic signals using impulses.

3) DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

4) Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

5) The inverse DTFT expresses $x[n]$ as a weighted "sum" of sinusoids of different frequencies.

6) The sum in the DTFT does not converge for all signals.

7) Notation $X(e^{j\omega})$ is used for later consistency with the z-transform.

8) DTFT pairs

a)

$$x[n] = \alpha^n u[n] \overset{DTFT}{\longleftrightarrow} X(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}}$$

b)

$$x[n] = \begin{cases} 1, & |n| \leq M \\ 0, & |n| > M \end{cases} \overset{DTFT}{\longleftrightarrow} X(e^{j\omega}) = \frac{\sin(\omega(2M+1)/2)}{\sin(\omega/2)}$$

c)

$$x[n] = \frac{\sin(Wn)}{\pi n} \overset{DTFT}{\longleftrightarrow} X(e^{j\omega}) = \begin{cases} 1, & |\omega| < W \\ 0, & W < |\omega| \leq \pi \end{cases}$$

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d)

$$x[n] = \delta[n] \overset{DTFT}{\longleftrightarrow} X(e^{j\omega}) = 1$$

e)

$$x[n] = \frac{1}{2\pi} \overset{DTFT}{\longleftrightarrow} X(e^{j\omega}) = \delta(\omega), \quad -\pi < \omega \leq \pi$$

9) The width of a signal in time is inversely related to the width of its DTFT.

Video Screenshots

- The DTFT is used for analysis of discrete-time signals
- The DTFT applies to discrete-time nonperiodic signals, but can be extended to periodic signals

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}\,d\omega \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

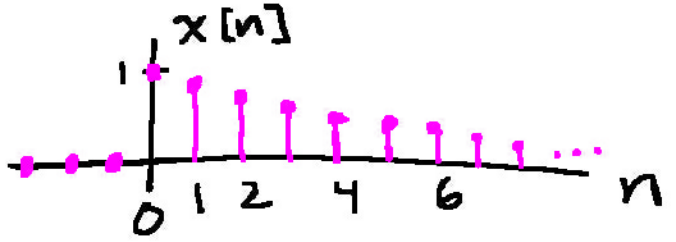
- Express $x[n]$ as weighted sum of sinusoids

$$x[n] \approx \sum_i X(e^{j\omega_i})\frac{\Delta\omega}{2\pi}\, e^{j\omega_i n}$$

- $X(e^{j\omega})$ notation for consistency with z-transform
- Convergence

Example: Exponential Sequence

$x[n] = \alpha^n u[n]$



$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \alpha^n u[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n$$

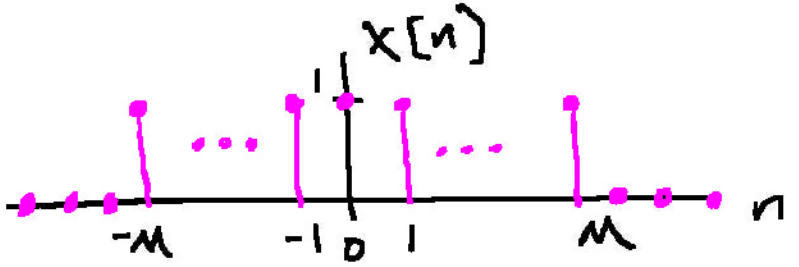
Geometric series

$$\sum_{n=0}^{\infty} \beta^n = \frac{1}{1-\beta}, \quad |\beta| < 1$$

$$X(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}}, \quad |\alpha| < 1$$

Example: Rectangular Pulse

$$x[n] = \begin{cases} 1 & |n| \le M \\ 0 & |n| > M \end{cases}$$



$$X(e^{j\omega}) = \sum_{n=-M}^{M} e^{-j\omega n} \qquad \text{let } m = n + M$$

$$= \sum_{m=0}^{2M} e^{-j\omega(m-M)}$$

$$= e^{j\omega M} \sum_{m=0}^{2M} e^{-j\omega m}$$

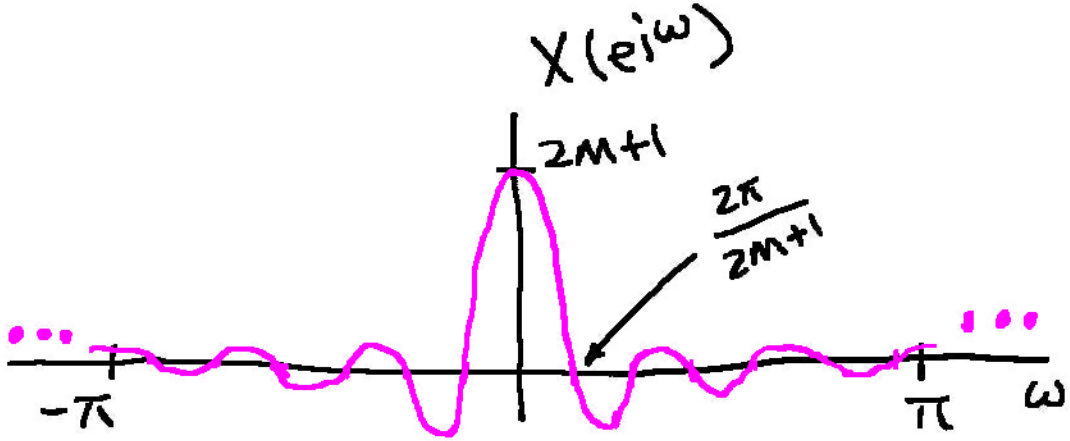
finite geometric series $\sum_{m=0}^{L} \beta^m = \frac{1-\beta^{L+1}}{1-\beta}, \beta \neq 1$

$$= e^{j\omega M} \frac{1 - e^{-j\omega(2M+1)}}{1 - e^{-j\omega}} \qquad (\omega \neq 0)$$

$\frac{e^{j\omega(2M+1)/2}}{e^{j\omega/2}}$

$$= \frac{e^{j\omega(2M+1)/2} - e^{-j\omega(2M+1)/2}}{e^{j\omega/2} - e^{-j\omega/2}} \qquad \frac{1}{2j}, \ \frac{1}{2j}$$

$$X(e^{j\omega}) = \frac{\sin(\omega(2M+1)/2)}{\sin(\omega/2)}$$



one period on $(-\pi, \pi]$

Example: Inverse DTFT of Rectangular Spectrum

$$X(e^{j\omega}) = \begin{cases} 1 & |\omega| < W \\ 0 & W < |\omega| < \pi \end{cases}$$

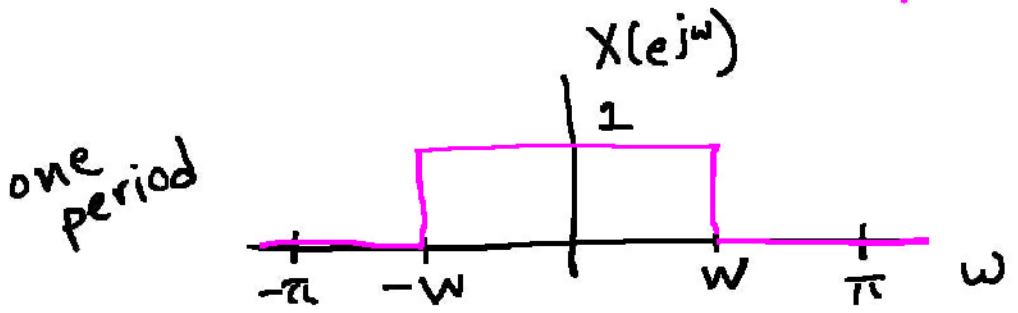
$$x[n] = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega n}\, d\omega$$

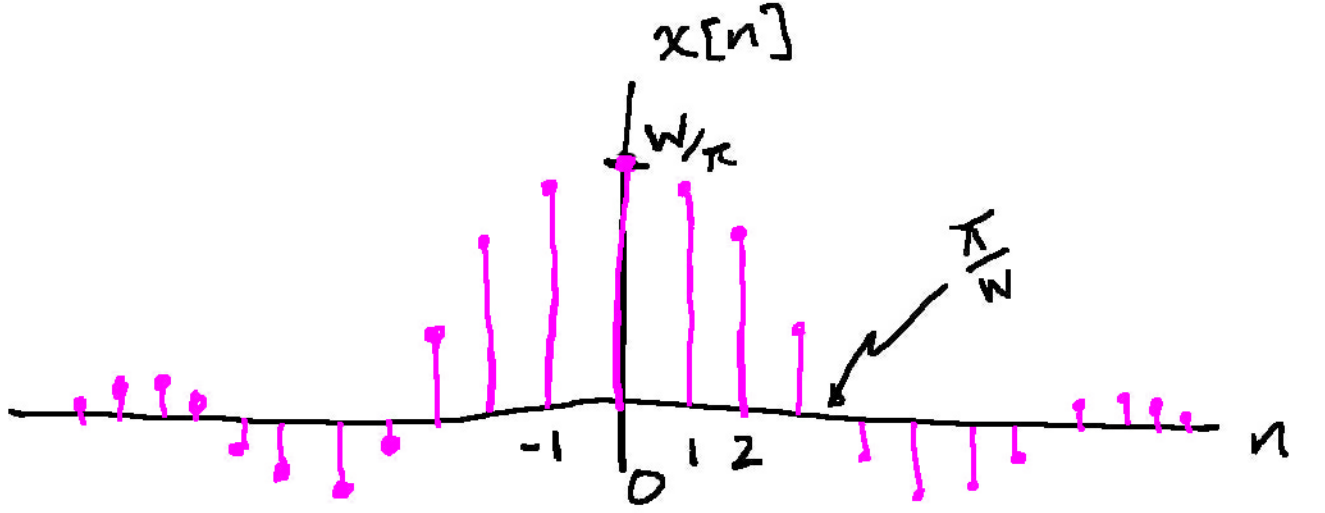
$$= \frac{1}{2\pi}\,\frac{1}{jn}\, e^{j\omega n}\Big|_{-W}^{W} \qquad n \neq 0$$

$$= \frac{1}{2\pi jn}\left(e^{jWn} - e^{-jWn}\right)$$

$$x[n] = \frac{\sin(Wn)}{\pi n}$$

note: $\lim_{n\to 0} \frac{\sin(Wn)}{\pi n} = \frac{W}{\pi}$

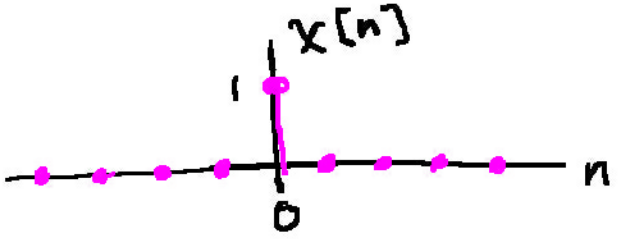




"width" in n and ω inversely related

Example: Impulse

$x[n] = \delta[n]$



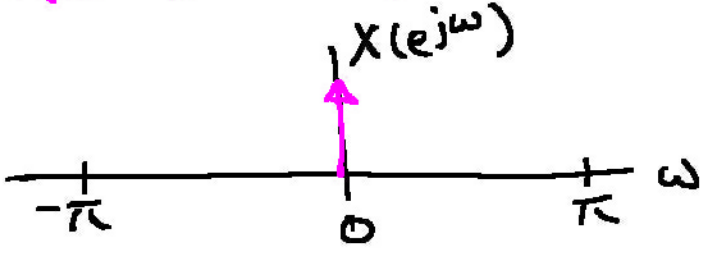
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n]\, e^{-j\omega n}$$

$$X(e^{j\omega}) = 1$$



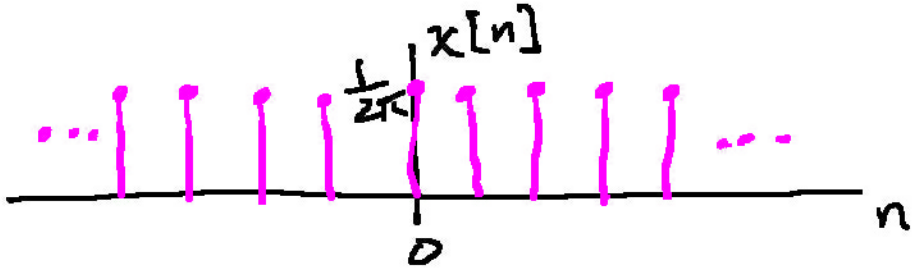
Example: Inverse DTFT of Impulse

$X(e^{j\omega}) = \delta(\omega) \quad -\pi < \omega \leq \pi$



$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(\omega)\, e^{j\omega n}\, d\omega$$

$$x[n] = \frac{1}{2\pi}$$



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